

spectrum agenda, the US government would give investors and technologists the confidence to devote resources to new ventures that make open spectrum a reality.

1. Fostering Effective Cooperation

The first step is to enhance existing unlicensed bands, which were not designed with open spectrum in mind. The FCC should work with the private sector and the technical community to identify minimal requirements to facilitate efficient spectrum sharing. In the near term, this could include service rules for the 5 GHz band to allow for continued growth of wireless data networking applications. These should not pre-determine technology or applications, but could include general requirements such as mandating that devices be capable of two-way packet-switched communications. The FCC should also identify restrictions in its existing rules, such as outmoded prohibitions on repeaters, that could be removed to allow for greater spectrum sharing.

In the future, as it establishes new unlicensed bands and eliminates underlay restrictions, the FCC could define additional “rules of the road,” either as requirements or as advisory “best practices.” For example, companies could be encouraged to build devices that modulate their output based on actual conditions, or that repeat traffic for other users, allowing for meshed architectures.

Whatever rules are adopted should be developed in consultation with industry representatives and technical experts to ensure they do not over- or under-specify standards. Reasonable accommodations should be made for uses of spectrum other than data networking, including scientific activity such as radio astronomy. Different rules may apply to particular bands or techniques. Whatever decisions are made will need to be reviewed periodically as conditions evolve.

2. Expanded Unlicensed Spectrum

Improving existing unlicensed bands isn’t enough. Most are so